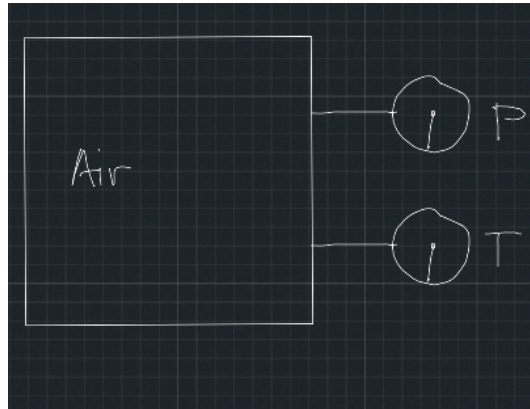


1 MC Uncertainty Propagation

For a data reduction equation (DRE) of the form:

$$r = f(x_1, x_2, \dots, x_L)$$

- Generate random values for each of the inputs
- For each set of random values, evaluate random
- This generates a set of values $r = \{r_1, r_2, \dots, r_m\}$
- Then evaluate the statistics of the set r



After you perform uncertainty analyses on the measured T and P value, calculate an uncertainty interval for density

$$\rho = \frac{P}{R_{air}T}$$

2 MC analysis

$$r = \rho$$

$$x_1 = p$$

$$x_2 = T$$

$$x_3 = R_{air}$$

Suppose:

$$\bar{T} = 298, U_t = 2k$$

$$T = 298 \pm 2k(95\%)$$

$$P = \bar{P} \pm U_p k(95\%)$$

$$P = 100 \pm 1k(95\%)$$

2.1 the code for this is:

```

m<-2e5 # of MC replicates

P <- numeric(M)

t <- numeric(M)

rho <- numeric(M)

Rair <- 287

for (m in 1:M) {
  p[m] <- rnorm(n=1, mean=1e5, sd = 500)
  t[m] <- rnorm(n=1, mean=298, sd = 1)
  rho[m] <- p[m]/Rair/t[m]
}
P<-0.95#conf.level
Pmin <- (1-p)/2
Pmax<-(1+p)/2
quantile(rho,probs=c(Pmin,Pmax))

```

The quantile function works on a list of numbers $\rho = \rho_1, \dots, \rho_M$ sorted from smallest to largest. For $P = 95\%$, the quantile function finds the ρ values that correspond to the upper and lower limits of 95% of the values.

```

x = {1,2,3, .. ,100}
quantile(x,probs = c(Pmin,Pmax))

```

```

Pmin=(1-p)/2 = 0.025
Pmax=(1+p)/2=0.975

```

3 Manufacturer's correlation

$$\dot{m} = c(\rho_o Z)^{1/2}$$

$$r = x_1(x_2 x_3)^{1/2}$$

Units	Quantity	Mean Value	$u = (s^2 + b^2)^{1/2}$
?	c	0.0504	0.0000005
kg/m^2	ρ_P	1.15	0.2
-	-	-	-

-Obstruction Meter Theory

$$Q_1 = K_e A_o \left(\frac{2 \Delta P}{\rho_p} \right)^{1/2} Y$$

$$\dot{m} = \rho_p Q_1 = K_e A_o \left(2 \rho_p \Delta P \right)^{1/2} Y$$

$$\dot{m} = f(K_e, A_o, \rho_p, \Delta P, Y)$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 x_1 x_2 x_3 x_4 x_5

4 Numerical Integration

$$\dot{m} = 2\pi \int_0^R \rho v r dr$$

$$\dot{m} = 2\pi \rho \int_0^R r v dr$$

Where: $v = \left(\frac{2\Delta P}{\rho_0} \right)^{1/2}$ Bernoulli Eqn.

$$\dot{m} = 2\pi \rho \int_0^R r \left(\frac{2\Delta P}{\rho_0} \right)^{1/2} dr$$

$$\dot{m} = 2\pi \rho \left(\frac{2}{\rho_0} \right)^{1/2} \int_0^R r \Delta P^{1/2} dr$$

$= 2\pi \rho_p \left(\frac{2}{\rho_p} \right)^{1/2} \int_0^R r (\Delta P)^{1/2} dr$
 $2\pi \rho_p^{3/2}$ Evaluate with trapz()

r (in)	ΔP (Pa)
0.0	54
0.5	56
...	...
2.9	30
3.0	0

4.0.1 algorithm

- generate a random value for ρ_o :
 $\rho_oD < - \text{rnorm}(n=1, \text{mean}=\bar{\rho}_0, \text{sd} = u_{\rho_0})$
- Generate one random value for each of the r values

```

r1<-rnorm(n=1,mean=rbar1, sd = ur1)
r2<-rnorm(n=1,mean=rbar2, sd = ur2)
  
```

- Generate one random value for each of the ΔP s
 $dP1 \leftarrow \text{rnorm}(n=1, \text{mean} = 54, \text{sd} = U_{\Delta P_1})$

- Define
 $r \leftarrow c(r1, r2, \dots)$
 $dP \leftarrow c(dP1, dP2, \dots)$
 $m \dot{\leftarrow} \frac{1}{2} \pi \text{ rdoD}^{1/2} * \text{trapez}(r, r * \text{sqr}(dP))$
 - Repeat M times, where $M = 2e5$
 - Save in each time
 - Use quantile on the saved in values.